

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

303000091 - Ampliación De Estadística Y Ciencia De Datos

DEGREE PROGRAMME

30AH - Máster Universitario En Matemáticas Avanzadas

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

Index

Learning guide

1. Description.....	1
2. Faculty.....	1
3. Prior knowledge recommended to take the subject.....	2
4. Skills and learning outcomes	2
5. Brief description of the subject and syllabus.....	4
6. Schedule.....	7
7. Activities and assessment criteria.....	10
8. Teaching resources.....	15
9. Other information.....	15

1. Description

1.1. Subject details

Name of the subject	303000091 - Ampliación de Estadística y Ciencia de Datos
No of credits	6 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 2
Tuition period	February-June
Tuition languages	English
Degree programme	30AH - Máster Universitario en Matemáticas Avanzadas
Centre	30 - Esc. Politéc. Enseñanza Superior (epes)
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Carlos Eduardo Gonzalez Guillen (Subject coordinator)	Matemáticas	carlos.gguillen@upm.es	M - 11:00 - 14:00 Tu - 11:00 - 14:00 Pedir cita por email

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

- Estadística Avanzada
- Machine Learning: Models And Applications

3.2. Other recommended learning outcomes

- python programming
- advanced programming skills

4. Skills and learning outcomes *

4.1. Skills to be learned

CO1 - Conocer los principales métodos y teorías matemáticas, próximos a la frontera del conocimiento, que permitan tener una amplia perspectiva en al menos tres de las siguientes áreas: Análisis Matemático, Métodos Numéricos, Ecuaciones Diferenciales, Estadística y Ciencia de Datos, Álgebra, Geometría y Topología. TIPO: Conocimientos o contenidos.

CO2 - Conocer demostraciones matemáticas avanzadas que permitan relacionar de una manera rigurosa las hipótesis con las conclusiones en al menos tres de las áreas relacionadas en el CO1. TIPO: Conocimientos o contenidos.

CO3 - Conocer diversas aplicaciones de las teorías matemáticas y su modelización en otros entornos o disciplinas que permitan adaptar los conocimientos matemáticos a nuevos problemas. TIPO: Conocimientos o contenidos.

CP1 - Utilizar programas informáticos especializados como laboratorio para diseñar, construir o validar hipótesis científicas en alguno de los ámbitos de las matemáticas. TIPO: Competencias.

CP2 - Ser capaz de trabajar en equipos multidisciplinares que involucren conocimientos matemáticos avanzados en áreas teóricas o aplicadas. TIPO: Competencias.

CP3 - Seleccionar y comprender la bibliografía científica sobre un tema concreto, extrayendo la información relevante. TIPO: Competencias.

CP4 - Comunicar de forma oral y escrita conocimientos y conclusiones científicas, valorando diferentes interpretaciones en públicos especializados y no especializados de una manera clara y rigurosa. TIPO: Competencias

CP5 - Complementar la formación matemática de manera autónoma en avances específicos de las Matemáticas, seleccionando la bibliografía adecuada y planteando cuestiones de interés. TIPO: Competencias

CP6 - Reflexionar sobre la relevancia social y ética de la aplicación de los conocimientos adquiridos a determinados problemas. TIPO: Competencias

HA1 - Aplicar los conocimientos matemáticos adquiridos a la resolución de problemas nuevos de una manera crítica, generando hipótesis de trabajo claras, definiciones precisas, nuevos enfoques con fundamento matemático y soluciones originales. TIPO: Habilidades o destrezas

HA2 - Ser capaz de abstraer y detectar las dificultades relevantes ante problemas complejos para poder plantearlos correctamente y resolverlos con técnicas matemáticas avanzadas. TIPO: Habilidades o destrezas

HA3 - Dominar el rigor matemático en las técnicas de demostración aprendidas para su posterior uso en el enunciado y demostraciones de nuevos teoremas. TIPO: Habilidades o destrezas

HA4 - Poseer destreza para proponer, validar e interpretar modelos matemáticos de diferente dificultad en situaciones reales complejas, utilizando las herramientas matemáticas más adecuadas a los fines que se persigan, y valorando las hipótesis y conclusiones asociadas a cada nivel de dificultad del modelo. TIPO: Habilidades o destrezas

HA5 - Adaptar teorías y técnicas matemáticas avanzadas al estudio de problemas en otros ámbitos con criterio científico para valorar su repercusión y formular las hipótesis adecuadas. TIPO: Habilidades o destrezas

HA6 - Interpretar adecuadamente la relevancia de nuevos resultados en Matemáticas, su aplicabilidad en situaciones particulares donde pudieran suponer nuevos avances y su relación con los resultados existentes. TIPO: Habilidades o destrezas

4.2. Learning outcomes

RA1 - To understand and apply the fundamental models in unsupervised, supervised, and reinforcement learning problems.

RA5 - Desarrollar trabajos autónomos o en grupo, integrando bibliografía científica y, en su caso, métodos computacionales para profundizar en los contenidos de la asignatura o abordar problemas complejos.

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

This course delves into advanced statistical and machine learning algorithms designed to handle large, complex datasets with specific characteristics. It focuses on unsupervised learning and deep learning techniques for regression, classification, and reconstruction, with special attention to real-world industrial applications.

The course emphasizes theoretical foundations while maintaining a strong practical component, enabling students to understand and implement modern, data-driven models.

Contents:

Advanced data science concepts for the development of generative models.

Advanced algorithms for large-scale data processing: deep autoencoders and clustering techniques.

Unsupervised learning algorithms for high-dimensional data or complex dynamics.

Deep learning methods: predictive models, classification, and reconstruction, including a comparison between classical and generative models.

Extension and generalization of deep learning models: Physics-Informed Neural Networks (PINNs).

5.2. Syllabus

1. Generative Models and Mathematical Foundations
2. Approximate Inference in Machine Learning
3. Modern Architectures: CNNs and GNNs
 - 3.1. Convolutional Networks: image and signal processing
 - 3.2. Architecture and training of modern CNNs
 - 3.3. Graph Neural Networks: structure and message passing
4. Deep Generative Models
 - 4.1. Normalizing Flows
 - 4.2. Variational Autoencoders (VAEs)
 - 4.3. Generative Adversarial Networks (GANs)
5. Representation Learning
 - 5.1. Latent representations
 - 5.2. Distributed encoding
 - 5.3. Pattern detection and compression
6. Topic 6: Introduction to Generative Language Models
 - 6.1. Transformer architecture
 - 6.2. Pretraining and scalability
7. Transfer and Multitask Learning
 - 7.1. Fine-tuning
 - 7.2. Domain transfer
 - 7.3. Multitask learning
8. Meta-Learning
 - 8.1. Optimization and adaptation
 - 8.2. Model-based and Bayesian meta-learning
9. Physics-Informed Neural Networks (PINNs)
 - 9.1. Motivation and mathematical formulation
 - 9.2. Incorporating differential equations into neural networks

9.3. Applications in physical modeling and computational engineering

10. Applications

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Tema 1 Duration: 02:00 Lecture Tema 2 Duration: 02:00 Lecture			
2	Tema 3.1 Duration: 02:00 Lecture Prácticas Duration: 02:00 Laboratory assignments			
3	Tema 3.2 Duration: 02:00 Lecture Tema 3.3 Duration: 02:00 Lecture			Entrega evaluable, practica o proyecto individual Individual work Progressive assessment and Global Examination Presential Duration: 00:00
4	Tema 4.1 Duration: 02:00 Lecture Prácticas Duration: 02:00 Laboratory assignments			
5	Tema 4.2 Duration: 02:00 Lecture Tema 4.3 Duration: 02:00 Lecture			
6	Tema 5.1 Duration: 02:00 Lecture Prácticas Duration: 02:00 Laboratory assignments			Entrega evaluable, practica o proyecto individual Individual work Progressive assessment and Global Examination Presential Duration: 00:00

7	Tema 5.2 Duration: 01:30 Lecture Tema 5.3 Duration: 01:30 Lecture Examen Teórico Duration: 01:00 Additional activities			Examen teórico Written test Progressive assessment Presential Duration: 01:00
8	Tema 6.1 Duration: 01:30 Lecture Prácticas Duration: 01:30 Laboratory assignments Presentación proyecto por equipos Duration: 01:00 Additional activities			Proyecto por equipos Group presentation Progressive assessment and Global Examination Presential Duration: 01:00
9	Tema 6.2 Duration: 02:00 Lecture Tema 7.1 Duration: 02:00 Lecture			Entrega evaluable, practica o proyecto individual Individual work Progressive assessment and Global Examination Presential Duration: 00:00
10	Tema 7.2 Duration: 02:00 Lecture Prácticas Duration: 02:00 Laboratory assignments			
11	Tema 7.3 Duration: 02:00 Lecture Tema 8.1 Duration: 02:00 Lecture			
12	Tema 8.2 Duration: 02:00 Lecture Prácticas Duration: 02:00 Laboratory assignments			Entrega evaluable, practica o proyecto individual Individual work Progressive assessment and Global Examination Presential Duration: 00:00
13	Tema 9.1 Duration: 02:00 Lecture Tema 9.2 Duration: 02:00 Lecture			

14	Tema 9.3 Duration: 01:30 Lecture Prácticas Duration: 01:30 Laboratory assignments Examen Teórico Duration: 01:00 Additional activities			Examen teórico Written test Progressive assessment Presential Duration: 01:00
15	Tema 10 Duration: 02:00 Lecture Tema 10 Duration: 02:00 Lecture			
16	Presentación proyecto por equipos Duration: 01:00 Additional activities			Proyecto por equipos Group presentation Progressive assessment and Global Examination Presential Duration: 01:00
17				Examen final Written test Global examination Presential Duration: 02:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
3	Entrega evaluable, practica o proyecto individual	Individual work	Face-to-face	00:00	10%	5 / 10	CO1 CO3 CP1 CP4 CP5 HA1 HA2 HA4 HA5 HA6
6	Entrega evaluable, practica o proyecto individual	Individual work	Face-to-face	00:00	10%	5 / 10	CO1 CO3 CP1 CP4 CP5 HA1 HA2 HA4 HA5 HA6
7	Examen teórico	Written test	Face-to-face	01:00	15%	5 / 10	CO2 CO3 HA3
8	Proyecto por equipos	Group presentation	Face-to-face	01:00	10%	5 / 10	CO1 CO3 CP1 CP2 CP3 CP4 CP5 CP6 HA1 HA2 HA4 HA5 HA6

9	Entrega evaluable, practica o proyecto individual	Individual work	Face-to-face	00:00	10%	5 / 10	CO1 CO3 CP1 CP4 CP5 HA1 HA2 HA4 HA5 HA6
12	Entrega evaluable, practica o proyecto individual	Individual work	Face-to-face	00:00	10%	5 / 10	CO1 CO3 CP1 CP4 CP5 HA1 HA2 HA4 HA5 HA6
14	Examen teórico	Written test	Face-to-face	01:00	15%	5 / 10	CO2 CO3 HA3
16	Proyecto por equipos	Group presentation	Face-to-face	01:00	10%	5 / 10	CO1 CO3 CP1 CP2 CP3 CP4 CP5 CP6 HA1 HA2 HA4 HA5 HA6

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
3	Entrega evaluable, practica o proyecto individual	Individual work	Face-to-face	00:00	10%	5 / 10	CO1 CO3 CP1 CP4 CP5 HA1 HA2 HA4 HA5 HA6

6	Entrega evaluable, practica o proyecto individual	Individual work	Face-to-face	00:00	10%	5 / 10	CO1 CO3 CP1 CP4 CP5 HA1 HA2 HA4 HA5 HA6
8	Proyecto por equipos	Group presentation	Face-to-face	01:00	10%	5 / 10	CO1 CO3 CP1 CP2 CP3 CP4 CP5 CP6 HA1 HA2 HA4 HA5 HA6
9	Entrega evaluable, practica o proyecto individual	Individual work	Face-to-face	00:00	10%	5 / 10	CO1 CO3 CP1 CP4 CP5 HA1 HA2 HA4 HA5 HA6
12	Entrega evaluable, practica o proyecto individual	Individual work	Face-to-face	00:00	10%	5 / 10	CO1 CO3 CP1 CP4 CP5 HA1 HA2 HA4 HA5 HA6
16	Proyecto por equipos	Group presentation	Face-to-face	01:00	10%	5 / 10	CO1 CO3 CP1 CP2 CP3 CP4 CP5 CP6 HA1 HA2

							HA4 HA5 HA6
17	Examen final	Written test	Face-to-face	02:00	40%	5 / 10	CO2 CO3 HA1 HA2 HA3 HA4 HA5

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Examen Extraordinario	Written test	Face-to-face	02:00	40%	10 / 10	CO2 CO3 HA1 HA2 HA3 HA4 HA5
Entrega evaluable, práctica y/o trabajo individual	Individual work	Face-to-face	00:00	40%	5 / 10	CO1 CO3 CP1 CP4 CP5 HA1 HA2 HA4 HA5 HA6
Proyecto por equipos	Group presentation	Face-to-face	00:30	20%	5 / 10	CO1 CO3 CP1 CP2 CP3 CP4 CP5 CP6 HA1 HA2 HA4 HA5 HA6

7.2. Assessment criteria

The primary goal of the evaluation is to verify students' ability to implement the methods taught in class. This includes programming the algorithms (directly or using libraries) and applying them to real-world or realistic datasets.

Evaluation components:

In-class evaluation: to encourage active participation rather than passive note-taking

Continuous evaluation: based on 4-5 short assessments throughout the semester to keep students up-to-date

Project evaluations: based on applications in scientific or engineering contexts, emphasizing practical application of classroom concepts

Each project will also involve a 5-10 minute in-class presentation, assessing both presentation skills and understanding. Participation in classmates' presentations will also be evaluated.

Weighting of Evaluation Components:

Attendance and participation: 10%

Assessable assignments, practicals, and individual projects: 40%

Theory exams: 30%

Team project evaluation: 20%

Extraordinary evaluation will also involve:

An assignment, practical, and/or individual projects proposed during the course: 40%

A team project evaluation: 20%

A Theory exam: 40%

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Slides - Theory notes	Bibliography	Teaching material, available in Moodle.
Python Programs	Bibliography	Application examples of Python algorithms, available in Moodle.
Moodle webpage	Web resource	Teaching material (notes, examples, programs) for students enrolled in the course is published on the website http://moodle.upm.es/titulaciones/oficiales .
Python	Equipment	Open source software

9. Other information

9.1. Other information about the subject

The 25/26 course is the first one in which this subject is taught. The schedule, syllabus and indications given in this guide are indicative and may be modified by the faculty during the course depending on the teaching needs and the needs of each evaluation activity.

In the practical computer work the students will be able to use, among others, the Python program and the specialized libraries of Machine Learning.

Annotated Bibliography

Recommended readings will include selected chapters from:

Deep Learning, by I. Goodfellow, Y. Bengio, and A. Courville

Deep Learning ? Foundations and Concepts, by Bishop and Bishop, a pedagogical text on deep learning

Probabilistic Machine Learning, by K. Murphy, covering nearly all topics from a probabilistic perspective, with extensive references to foundational research papers

Reinforcement Learning: An Introduction, by Sutton and Barto ? a classic for understanding the foundations of reinforcement learning

Deep Learning with PyTorch, by Stevens, Antiga, and Viehmann, and <https://pytorch.org/tutorials/> ? both useful resources for learning PyTorch

Recent papers from JMLR, IEEE TNNLS